

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0683

Nrg Rockford I 230 kV
May 28, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0683. The Interconnection Customer proposes a 144.1 MW solar photovoltaic facility interconnecting at the Nrg Rockford I 230 kV substation in the Illinois area. The request was received on October 01, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 263 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,020,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
P.T.S-Medora 345.0 kV	Allegheny Energy Units 3 4 & 5-Sugar Hill 345.0 kV	101.7	1,619	7.6
Pine Creek Project Substation-Handsome Lake 500.0 kV	Piney Creek Project-Lions Rd 500.0 kV	100.9	2,282	12.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
P.T.S-Medora 345.0 kV	Reconductor line	\$9,298,000	2% (\$206,000)
Piney Crk Project-Handsome Lake 500.0 kV	Reconductor line	\$36,746,000	2% (\$814,000)
	Total allocated to project		\$1,020,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.